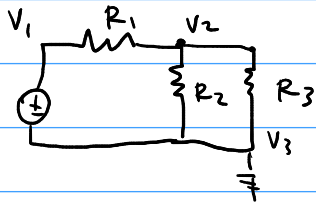


Numerically Solving Circuits with Gradient Descent and Power Minimization!

How to numerically solve circuit with optimization techniques!

Consider ckt. :



$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{R_1} & -\frac{1}{R_1} & 0 \\ 0 & \frac{1}{R_2} & -\frac{1}{R_2} \\ 0 & \frac{1}{R_3} & -\frac{1}{R_3} \end{bmatrix}}_G \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

with V_1 = fixed voltage and $V_3 = \text{gnd} = 0 \text{ V}$.

Easy to solve by hand, but how can we make ckt. solver?

Consider power minimization:

If voltage updates to the configuration with the smallest power dissipation we have a voltage going "downhill" along the power gradient:

$$V(t + \Delta t) = V(t) - \nabla_V P \Delta t$$

Voltage at the next moment in time, moves to the direction of smaller power = { GRADIENT DESCENT }

$$\boxed{\frac{dV}{dt} = - \frac{dP}{dV}}$$

So let's find the gradient!

first we need expression for total power.

$$P = \sum_k I_k^2 R_k \quad \leftarrow \text{power is sum of power through each resistor } P_k = I_k^2 R_k$$

$$\text{w/ } I_k = G_k \cdot \vec{V}$$

$\uparrow$
kth row of G

$$\vec{I} = G \vec{V} \quad , \quad \sum_k I_k^2 R_k = \sum_k I_k R_k I_k = \vec{I}^T \underset{\substack{\uparrow \\ \text{diagonal matrix}}}{R} \vec{I}$$

$$P = \vec{V}^T G^T R G \vec{V}$$

So P gives total power and V's should update to minimize this $\rightarrow$ gradient descent.

currents & voltages should "work" to minimize power dissipation.

$$\frac{\partial P}{\partial \vec{V}} = \frac{\partial}{\partial \vec{V}} \vec{V}^T \underbrace{G^T R G}_A \vec{V} = (A^T + A) \vec{V}$$

$\nearrow$ see matrix calculus notes for derivation / explanation.

$$\boxed{\frac{\partial P}{\partial \vec{V}} = 2G^T R G \vec{V}}$$

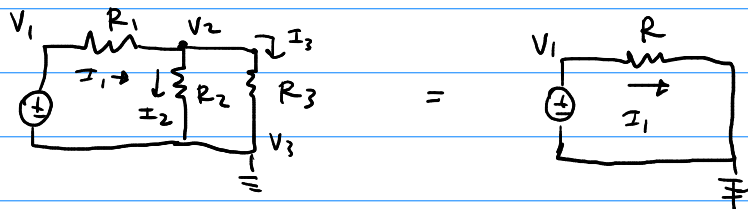
$$\boxed{\frac{\partial \vec{V}}{\partial t} = -2G^T R G \vec{V}}$$

So all we need to do is time step $\vec{v}$ w/ $\frac{\partial \vec{v}}{\partial t}$

and fix any fixed nodes on each time step.

Let's see if it works by doing analytical calculation and comparing numerical solution.

Analytical:



$$R = R_1 + \frac{R_2 \cdot R_3}{R_2 + R_3}$$

$$I_1 = \frac{V_1}{R}, \quad I_2 = \frac{V_2}{R_2}, \quad I_3 = \frac{V_3}{R_3}$$

$$V_2 = V_{R_2} = V_{R_3}, \quad V_1 - V_2 = I_1 \cdot R_1 \rightarrow V_2 = V_1 - I_1 R_1$$

$$I_2 = \frac{V_2}{R_2}, \quad I_3 = \frac{V_2}{R_3}$$

$$I_3 = \frac{I_1 R_2}{R_2 + R_3}$$

$$I_2 = I_1 - I_3$$

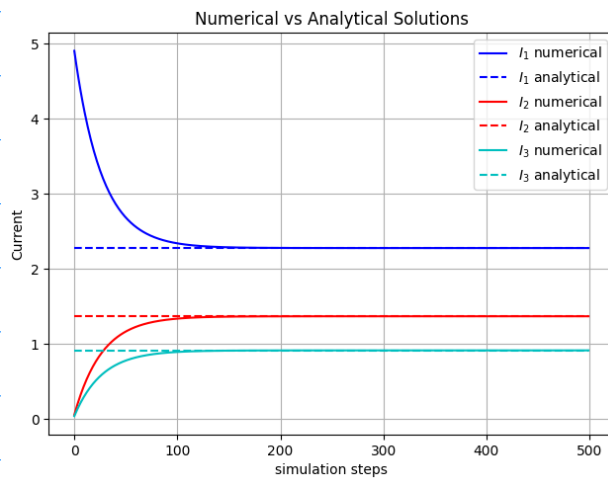
This is the essential code. w/ $A = 2G^T RT$

```
# voltage vector to be time stepped
v = np.array([V1, V2, V3]).reshape(3,1)

for i in range(nSteps):
    v = v - dt * A @ v
    v[0] = V1
    v[2] = V3

    vHist.append(v.ravel() * 1)
```

Results:



It works!